Tutorial 1

1. The following is a sample image of 5x5 pixels

23	14	179	243	233
16	234	155	177	2
14	234	12	28	173
200	149	176	20	17
73	1	188	39	208

A 2x2 template is given as below

12	28
175	20

(a) Derive the equation of cross correlation (or ZNCC).
(b) Use the cross correlation technique to locate the most likely match in the sample image and give its matching score.
(c) Repeat (b) with SAD.

2. What are the major steps in PCA based face recognition?

3.

For the following square matrix:

$$\begin{pmatrix} 3 & 0 & 1 \\ -4 & 1 & 2 \\ -6 & 0 & -2 \end{pmatrix}$$

Decide which, if any, of the following vectors are eigenvectors of that matrix and give the corresponding eigenvalue.

$$\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

Find all eigen values, and eigen vectors.

4.

- What do the eigenvectors of the covariance matrix give us?
- At what point in the PCA process can we decide to compress the data? What effect does this have?

5. There are 2 clusters of 2-dimensional data. All example points are located at integer position.

   Cluster-1: (5,9), (6,10),(7,10), (7,9)
   Cluster-2: (17, 12), (18, 13), (19, 12), (19,14), (20,13)

   (1) Derive the cluster centers.
   (2) A test data is given as (10, 10). Determine which cluster it belongs to, using the NCC (Nearest Center Classifier).
   (3) Apply the PCA (Principal Component Analysis) to reduce the data into one dimension.
   (4) Derive the cluster centers in one dimension.
   (5) Transform the test data (10, 10) into the reduced data dimension and determine which cluster it belongs to.